

Lecture 02 - Rendering Pipeline

This lecture turns the vague idea of rendering into a concrete sequence. It is the most important debugging map in the course: if an image is wrong, the pipeline tells you where the error could have entered.

Source chunk	course_text_parts/03_lectures/02_rendering-pipeline.txt
Extracted slide pages	94
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Per-slide commentary

Each extracted slide page gets its own note. Use this section as the spoken commentary that the terse slide deck is missing.

Page 1	Untitled slide
Source cue	No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content.
Professor-style explanation	The slide 'Untitled slide' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content. The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Untitled slide. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Untitled slide' into a causal sentence instead of repeating the slide title?
Page 2	Object-based Rendering
Source cue	- Image generation done through rasterization / - Scene objects formed by 3D models (primitives) / are transformed through successive coordinate systems / - Primitives are rasterized into fragments, which are pixel predecessors / - A subset of the fragments becomes pixels

Professor-style explanation	The slide 'Object-based Rendering' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: - Image generation done through rasterization / - Scene objects formed by 3D models (primitives) / are transformed through successive coordinate systems / - Primitives are rasterized into fragments, which are pixel predecessors / - A subset of the fragments becomes pixels. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about Object-based Rendering. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: - Image generation done through rasterization / - Scene objects formed by 3D models (primitives) / are transformed through successive coordinate systems / - Primitives are rasterized into fragments, which are pixel predecessors / - A subset of the fragments becomes pixels
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'Object-based Rendering' in the rendering pipeline?

Page 3	2.1 Rendering Process
Source cue	2.2 OpenGL Overview / 2.3 OpenGL Rendering / 2.4 OpenGL Rendering Pipeline / 2.5 Fragment Tests and Operations / 2.6 Framebuffer
Professor-style explanation	The slide '2.1 Rendering Process' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: 2.2 OpenGL Overview / 2.3 OpenGL Rendering / 2.4 OpenGL Rendering Pipeline / 2.5 Fragment Tests and Operations / 2.6 Framebuffer. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about 2.1 Rendering Process. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: 2.2 OpenGL Overview / 2.3 OpenGL Rendering / 2.4 OpenGL Rendering Pipeline / 2.5 Fragment Tests and Operations / 2.6 Framebuffer
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for '2.1 Rendering Process' in the rendering pipeline?

Page 4	2.1 Rendering Process
Source cue	From 3D models over fragments to pixels

Professor-style explanation	The slide '2.1 Rendering Process' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: From 3D models over fragments to pixels. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about 2.1 Rendering Process. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: From 3D models over fragments to pixels
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for '2.1 Rendering Process' in the rendering pipeline?

Page 5	Geometry-based Rendering 1/2
Source cue	CPU Application GPU Rendering Display Hardware / Rendering / 3D Model Framebuffer / Process / Video
Professor-style explanation	The slide 'Geometry-based Rendering 1/2' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: CPU Application GPU Rendering Display Hardware / Rendering / 3D Model Framebuffer / Process / Video. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about Geometry-based Rendering 1/2. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: CPU Application GPU Rendering Display Hardware / Rendering / 3D Model Framebuffer / Process / Video
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'Geometry-based Rendering 1/2' in the rendering pipeline?

Page 6	Geometry-based Rendering 2/2
Source cue	- The rendering process is realized through the rendering pipeline / - Renders 3D models to 2D images / - Conceptually, the rendering pipeline consists of two GPU stages / - Geometry stage – processes geometric primitives / - Rasterization stage – processes fragments and pixels

Professor-style explanation	The slide 'Geometry-based Rendering 2/2' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: - The rendering process is realized through the rendering pipeline / - Renders 3D models to 2D images / - Conceptually, the rendering pipeline consists of two GPU stages / - Geometry stage – processes geometric primitives / - Rasterization stage – processes fragments and pixels. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about Geometry-based Rendering 2/2. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: - The rendering process is realized through the rendering pipeline / - Renders 3D models to 2D images / - Conceptually, the rendering pipeline consists of two GPU stages / - Geometry stage – processes geometric primitives / - Rasterization stage – processes fragments and pixels
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'Geometry-based Rendering 2/2' in the rendering pipeline?

Page 7	Application Stage
Source cue	- Specification of the model elements and the scene / as well as interaction capabilities / - Scene modeling: arrangement and attribution of the scene objects / - Calculation of animation paths / - Collision detection
Professor-style explanation	The slide 'Application Stage' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: - Specification of the model elements and the scene / as well as interaction capabilities / - Scene modeling: arrangement and attribution of the scene objects / - Calculation of animation paths / - Collision detection. The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Application Stage. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: - Specification of the model elements and the scene / as well as interaction capabilities / - Scene modeling: arrangement and attribution of the scene objects / - Calculation of animation paths / - Collision detection
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Application Stage' into a causal sentence instead of repeating the slide title?

Page 8	Geometry Stage
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Source cue	x x / - Geometric 3D transformation / Model Coordinate System World Coordinate System / - Alignment of individual scene objects in the overall scene / - Basic transformations: translation, scaling, rotation
Professor-style explanation	The slide 'Geometry Stage' explains how geometric objects change coordinate systems. A point, vector, normal, or local coordinate frame is multiplied by a transformation matrix or affected by an affine operation. Translations move positions, rotations change orientation, scaling changes size, and composed matrices combine several such effects. On the slide, the concrete items are: x x / - Geometric 3D transformation / Model Coordinate System World Coordinate System / - Alignment of individual scene objects in the overall scene / - Basic transformations: translation, scaling, rotation. The object-level relation is input coordinate space, transformation object, and output coordinate space.
Technical commentary	This slide is about Geometry Stage. The transformation objects on the slide move points, vectors, or coordinate frames from one space into another using matrices or affine operations. Concrete items shown: x x / - Geometric 3D transformation / Model Coordinate System World Coordinate System / - Alignment of individual scene objects in the overall scene / - Basic transformations: translation, scaling, rotation
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Geometry Stage'?
Page 9	Rasterization Stage
Source cue	- Conversion of primitives into fragments (rasterization) / - Discretization takes place in relation to a target grid / - Rasterization algorithms optimized for certain primitives / (e.g., lines, triangles) / - After rasterization per-fragment tests are performed
Professor-style explanation	The slide 'Rasterization Stage' explains the conversion from continuous geometry to a discrete sample grid. A mathematical line, triangle, or region is tested against pixel/sample positions. Covered samples become fragments, and attributes such as depth, color, normals, or texture coordinates can be interpolated across the primitive. On the slide, the concrete items are: - Conversion of primitives into fragments (rasterization) / - Discretization takes place in relation to a target grid / - Rasterization algorithms optimized for certain primitives / (e.g., lines, triangles) / - After rasterization per-fragment tests are performed. Rasterization creates fragment candidates; later tests decide whether those candidates become visible pixel updates.
Technical commentary	This slide is about Rasterization Stage. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - Conversion of primitives into fragments (rasterization) / - Discretization takes place in relation to a target grid / - Rasterization algorithms optimized for certain primitives / (e.g., lines, triangles) / - After rasterization per-fragment tests are performed
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.

Check yourself	Can you explain which samples/fragments are generated by 'Rasterization Stage'?
Page 10	2.2 OpenGL Overview
Source cue	Library for real-time computer graphics
Professor-style explanation	The slide '2.2 OpenGL Overview' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: Library for real-time computer graphics. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about 2.2 OpenGL Overview. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: Library for real-time computer graphics
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '2.2 OpenGL Overview'?
Page 11	OpenGL History
Source cue	- Unified graphics software interface was missing / as knowledge channeled into proprietary software packages / - HP's starbase / - SGI's Graphics Library (GL) / - Some standardization efforts existed
Professor-style explanation	The slide 'OpenGL History' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Unified graphics software interface was missing / as knowledge channeled into proprietary software packages / - HP's starbase / - SGI's Graphics Library (GL) / - Some standardization efforts existed. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about OpenGL History. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Unified graphics software interface was missing / as knowledge channeled into proprietary software packages / - HP's starbase / - SGI's Graphics Library (GL) / - Some standardization efforts existed
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL History'?
Page 12	OpenGL

Source cue	- OpenGL (Open Graphics Library) / - Platform independent 2D and 3D graphics API / - API specification is provided by the Khronos Group / - Technical implementation / - Rendering pipeline consisting of static and programmable pipeline stages
Professor-style explanation	The slide 'OpenGL' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: - OpenGL (Open Graphics Library) / - Platform independent 2D and 3D graphics API / - API specification is provided by the Khronos Group / - Technical implementation / - Rendering pipeline consisting of static and programmable pipeline stages. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about OpenGL. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: - OpenGL (Open Graphics Library) / - Platform independent 2D and 3D graphics API / - API specification is provided by the Khronos Group / - Technical implementation / - Rendering pipeline consisting of static and programmable pipeline stages
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'OpenGL' in the rendering pipeline?

Page 13	OpenGL vs. Vulkan
Source cue	- How about Vulkan? / - More modern than OpenGL / - Designed for high performance and low-level GPU control / - Better suited for complex, large-scale rendering engines / - Vulkan also harder to learn and requires more setup/code
Professor-style explanation	The slide 'OpenGL vs. Vulkan' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - How about Vulkan? / - More modern than OpenGL / - Designed for high performance and low-level GPU control / - Better suited for complex, large-scale rendering engines / - Vulkan also harder to learn and requires more setup/code. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about OpenGL vs. Vulkan. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - How about Vulkan? / - More modern than OpenGL / - Designed for high performance and low-level GPU control / - Better suited for complex, large-scale rendering engines / - Vulkan also harder to learn and requires more setup/code

Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL vs. Vulkan'?
Page 14	OpenGL Libraries
Source cue	- Different tastes of OpenGL / - OpenGL / - OpenGL ES - limited version for embedded systems / - WebGL - Javascript connection for Web Development / (functionality similar to OpenGL ES)
Professor-style explanation	The slide 'OpenGL Libraries' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Different tastes of OpenGL / - OpenGL / - OpenGL ES - limited version for embedded systems / - WebGL - Javascript connection for Web Development / (functionality similar to OpenGL ES). The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about OpenGL Libraries. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Different tastes of OpenGL / - OpenGL / - OpenGL ES - limited version for embedded systems / - WebGL - Javascript connection for Web Development / (functionality similar to OpenGL ES)
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Libraries'?
Page 15	OpenGL Functionality
Source cue	- Functional API / - Objects, scenes and images are described by a sequence of commands / (functional approach, low-level) / - Contrast: scene or object description (declarative approach, high-level) / - OpenGL context
Professor-style explanation	The slide 'OpenGL Functionality' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Functional API / - Objects, scenes and images are described by a sequence of commands / (functional approach, low-level) / - Contrast: scene or object description (declarative approach, high-level) / - OpenGL context. The concrete relation is between API state, GPU resource, shader input, and final rendering result.

Technical commentary	This slide is about OpenGL Functionality. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Functional API / - Objects, scenes and images are described by a sequence of commands / (functional approach, low-level) / - Contrast: scene or object description (declarative approach, high-level) / - OpenGL context
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Functionality'?

Page 16	OpenGL Context
Source cue	- Context is created and modified by the application, / and contains all the state variables of the OpenGL engine / - Any number of contexts per application possible, / but at most one active context per application / - Context is generally tied to an output window
Professor-style explanation	The slide 'OpenGL Context' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Context is created and modified by the application, / and contains all the state variables of the OpenGL engine / - Any number of contexts per application possible, / but at most one active context per application / - Context is generally tied to an output window. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about OpenGL Context. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Context is created and modified by the application, / and contains all the state variables of the OpenGL engine / - Any number of contexts per application possible, / but at most one active context per application / - Context is generally tied to an output window
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Context'?

Page 17	OpenGL Rendering Process
Source cue	- Initiation from the developer's perspective / - Creation of the output window / - Creation of the context / - Specification of the scene geometry / - Uploading of the scene geometry to the GPU

Professor-style explanation	The slide 'OpenGL Rendering Process' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: - Initiation from the developer's perspective / - Creation of the output window / - Creation of the context / - Specification of the scene geometry / - Uploading of the scene geometry to the GPU. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about OpenGL Rendering Process. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: - Initiation from the developer's perspective / - Creation of the output window / - Creation of the context / - Specification of the scene geometry / - Uploading of the scene geometry to the GPU
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'OpenGL Rendering Process' in the rendering pipeline?
Page 18	Window Creation with GLFW 1/2
Source cue	<pre>#include <GLFW/glfw3.h> / #include <stdlib.h> / #include <stdio.h> / static void error_callback(int error, const char* description) { / fputs(description, stderr);</pre>
Professor-style explanation	The slide 'Window Creation with GLFW 1/2' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: #include <GLFW/glfw3.h> / #include <stdlib.h> / #include <stdio.h> / static void error_callback(int error, const char* description) { / fputs(description, stderr). The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Window Creation with GLFW 1/2. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: #include <GLFW/glfw3.h> / #include <stdlib.h> / #include <stdio.h> / static void error_callback(int error, const char* description) { / fputs(description, stderr);
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Window Creation with GLFW 1/2' into a causal sentence instead of repeating the slide title?
Page 19	Window Creation with GLFW 2/2
Source cue	<pre>int main(void){ / GLFWwindow* window; / glfwSetErrorCallback(error_callback); / if (!glfwInit()) exit(EXIT_FAILURE); / window = glfwCreateWindow(640, 480, "Simple example", NULL, NULL);</pre>

Professor-style explanation	The slide 'Window Creation with GLFW 2/2' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: <code>int main(void){ / GLFWwindow* window; / glfwSetErrorCallback(error_callback); / if (!glfwInit()) exit(EXIT_FAILURE); / window = glfwCreateWindow(640, 480, "Simple example", NULL, NULL)</code> . The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Window Creation with GLFW 2/2. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: <code>int main(void){ / GLFWwindow* window; / glfwSetErrorCallback(error_callback); / if (!glfwInit()) exit(EXIT_FAILURE); / window = glfwCreateWindow(640, 480, "Simple example", NULL, NULL);</code>
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Window Creation with GLFW 2/2' into a causal sentence instead of repeating the slide title?

Page 20	OpenGL Naming Conventions
Source cue	- Function-Prefix "gl" / - <code>glClear</code> , ... / - Constant-Prefix "GL" / - <code>GL_POLYGON</code> , <code>GL_LINES</code> , ... / - Parameter types are explicitly encoded in function names
Professor-style explanation	The slide 'OpenGL Naming Conventions' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Function-Prefix "gl" / - <code>glClear</code> , ... / - Constant-Prefix "GL" / - <code>GL_POLYGON</code> , <code>GL_LINES</code> , ... / - Parameter types are explicitly encoded in function names. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about OpenGL Naming Conventions. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Function-Prefix "gl" / - <code>glClear</code> , ... / - Constant-Prefix "GL" / - <code>GL_POLYGON</code> , <code>GL_LINES</code> , ... / - Parameter types are explicitly encoded in function names
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Naming Conventions'?
Page 21	2.3 OpenGL Rendering
Source cue	Transfer of 3D models to the GPU

Professor-style explanation	The slide '2.3 OpenGL Rendering' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: Transfer of 3D models to the GPU. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about 2.3 OpenGL Rendering. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: Transfer of 3D models to the GPU
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '2.3 OpenGL Rendering'?

Page 22	OpenGL Rendering (until V2.1)
Source cue	- Rendering is done through Immediate Mode / void renderScene() { / glColor3f(0.0f, 1.0f, 0.0f); // drawing color becomes green / glBegin(GL_POLYGON); // begin primitive / glVertex2f(-0.5f,-0.5f); // 1. vertex
Professor-style explanation	The slide 'OpenGL Rendering (until V2.1)' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Rendering is done through Immediate Mode / void renderScene() { / glColor3f(0.0f, 1.0f, 0.0f); // drawing color becomes green / glBegin(GL_POLYGON); // begin primitive / glVertex2f(-0.5f,-0.5f); // 1. vertex. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about OpenGL Rendering (until V2.1). The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Rendering is done through Immediate Mode / void renderScene() { / glColor3f(0.0f, 1.0f, 0.0f); // drawing color becomes green / glBegin(GL_POLYGON); // begin primitive / glVertex2f(-0.5f,-0.5f); // 1. vertex
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Rendering (until V2.1)'?

Page 23	OpenGL Rendering (since V3.0)
Source cue	- Using Vertex Array Objects (VAOs) / - Vertex specification in CPU memory / - Vertex data transfer from CPU to GPU / - Memory layout description of the vertex data / - Number of vertices, number of components (x, y, z, ...)

Professor-style explanation	The slide 'OpenGL Rendering (since V3.0)' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Using Vertex Array Objects (VAOs) / - Vertex specification in CPU memory / - Vertex data transfer from CPU to GPU / - Memory layout description of the vertex data / - Number of vertices, number of components (x, y, z, ...). The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about OpenGL Rendering (since V3.0). The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Using Vertex Array Objects (VAOs) / - Vertex specification in CPU memory / - Vertex data transfer from CPU to GPU / - Memory layout description of the vertex data / - Number of vertices, number of components (x, y, z, ...)
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Rendering (since V3.0)'?

Page 24	1. Vertex Specification
Source cue	- Vertex data is specified as a C array // vertices of two triangles / GLfloat vertices[6][2] = { // 6 vertices with two components each (x,y) / {0.0, 1.0}, {1.0, 1.0}, {1.0, 0.0}, // 1. triangle / {0.5, 0.5}, {1.0, 0.3}, {0.7, 1.0} // 2. triangle
Professor-style explanation	The slide '1. Vertex Specification' explains the conversion from continuous geometry to a discrete sample grid. A mathematical line, triangle, or region is tested against pixel/sample positions. Covered samples become fragments, and attributes such as depth, color, normals, or texture coordinates can be interpolated across the primitive. On the slide, the concrete items are: - Vertex data is specified as a C array // vertices of two triangles / GLfloat vertices[6][2] = { // 6 vertices with two components each (x,y) / {0.0, 1.0}, {1.0, 1.0}, {1.0, 0.0}, // 1. triangle / {0.5, 0.5}, {1.0, 0.3}, {0.7, 1.0} // 2. triangle. Rasterization creates fragment candidates; later tests decide whether those candidates become visible pixel updates.
Technical commentary	This slide is about 1. Vertex Specification. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - Vertex data is specified as a C array // vertices of two triangles / GLfloat vertices[6][2] = { // 6 vertices with two components each (x,y) / {0.0, 1.0}, {1.0, 1.0}, {1.0, 0.0}, // 1. triangle / {0.5, 0.5}, {1.0, 0.3}, {0.7, 1.0} // 2. triangle
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by '1. Vertex Specification'?

Page 25	2. Vertex Data Transfer
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Source cue	- VBO allocation and data transfer via / - glBufferData(GL_ARRAY_BUFFER, size, data, GL_STATIC_DRAW); / - data – address in main memory (i.e., vertices or &vertices[0]) / - size – size of data in number of bytes / (i.e. sizeof(vertices) or numVertices*sizeof(GLfloat)*2)
Professor-style explanation	The slide '2. Vertex Data Transfer' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - VBO allocation and data transfer via / - glBufferData(GL_ARRAY_BUFFER, size, data, GL_STATIC_DRAW); / - data – address in main memory (i.e., vertices or &vertices[0]) / - size – size of data in number of bytes / (i.e. sizeof(vertices) or numVertices*sizeof(GLfloat)*2). The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about 2. Vertex Data Transfer. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - VBO allocation and data transfer via / - glBufferData(GL_ARRAY_BUFFER, size, data, GL_STATIC_DRAW); / - data – address in main memory (i.e., vertices or &vertices[0]) / - size – size of data in number of bytes / (i.e. sizeof(vertices) or numVertices*sizeof(GLfloat)*2)
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '2. Vertex Data Transfer'?
Page 26	3. Memory Layout Description
Source cue	- Memory layout is described by the following function / - glVertexAttribPointer(attribIndex, size, type, normalized, stride, offset); / - attribIndex – attribute number (0 for position, later more ...) / - size – number of components per vertex (2 for xy, 3 for xyz) / - type – base type (typically GL_FLOAT or GL_DOUBLE)
Professor-style explanation	The slide '3. Memory Layout Description' explains local illumination at a surface point. The surface normal defines orientation, the light vector defines incoming light, the view vector defines the observer, and material parameters scale ambient, diffuse, or specular terms. On the slide, the concrete items are: - Memory layout is described by the following function / - glVertexAttribPointer(attribIndex, size, type, normalized, stride, offset); / - attribIndex – attribute number (0 for position, later more ...) / - size – number of components per vertex (2 for xy, 3 for xyz) / - type – base type (typically GL_FLOAT or GL_DOUBLE). The object-level relation is light source, surface point, material response, and computed color.

Technical commentary	This slide is about 3. Memory Layout Description. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Memory layout is described by the following function / - <code>glVertexAttribPointer(attribIndex, size, type, normalized, stride, offset);</code> / - <code>attribIndex</code> – attribute number (0 for position, later more ...) / - <code>size</code> – number of components per vertex (2 for xy, 3 for xyz) / - <code>type</code> – base type (typically <code>GL_FLOAT</code> or <code>GL_DOUBLE</code>)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '3. Memory Layout Description'?

Page 27	4. VAO Drawing 1/2
Source cue	- Drawing is done with the command / - <code>glDrawArrays(mode, first, count);</code> / - <code>mode</code> – specifies the primitive type (e.g., <code>GL_TRIANGLES</code> or <code>GL_LINES</code>) / - From all vertices, the next count vertices are drawn from index first / - Actual vertex data no longer needs to be specified
Professor-style explanation	The slide '4. VAO Drawing 1/2' explains the conversion from continuous geometry to a discrete sample grid. A mathematical line, triangle, or region is tested against pixel/sample positions. Covered samples become fragments, and attributes such as depth, color, normals, or texture coordinates can be interpolated across the primitive. On the slide, the concrete items are: - Drawing is done with the command / - <code>glDrawArrays(mode, first, count);</code> / - <code>mode</code> – specifies the primitive type (e.g., <code>GL_TRIANGLES</code> or <code>GL_LINES</code>) / - From all vertices, the next count vertices are drawn from index first / - Actual vertex data no longer needs to be specified. Rasterization creates fragment candidates; later tests decide whether those candidates become visible pixel updates.
Technical commentary	This slide is about 4. VAO Drawing 1/2. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - Drawing is done with the command / - <code>glDrawArrays(mode, first, count);</code> / - <code>mode</code> – specifies the primitive type (e.g., <code>GL_TRIANGLES</code> or <code>GL_LINES</code>) / - From all vertices, the next count vertices are drawn from index first / - Actual vertex data no longer needs to be specified
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by '4. VAO Drawing 1/2'?

Page 28	4. VAO Drawing 2/2
Source cue	- There is a small number of primitive types / - Points / - Lines, Line Strips, Line Loops / - Triangles, Triangle Strips, Triangle Fans, Patches / v2 v2 v2

Professor-style explanation	The slide '4. VAO Drawing 2/2' explains the conversion from continuous geometry to a discrete sample grid. A mathematical line, triangle, or region is tested against pixel/sample positions. Covered samples become fragments, and attributes such as depth, color, normals, or texture coordinates can be interpolated across the primitive. On the slide, the concrete items are: - There is a small number of primitive types / - Points / - Lines, Line Strips, Line Loops / - Triangles, Triangle Strips, Triangle Fans, Patches / v2 v2 v2. Rasterization creates fragment candidates; later tests decide whether those candidates become visible pixel updates.
Technical commentary	This slide is about 4. VAO Drawing 2/2. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - There is a small number of primitive types / - Points / - Lines, Line Strips, Line Loops / - Triangles, Triangle Strips, Triangle Fans, Patches / v2 v2 v2
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by '4. VAO Drawing 2/2'?

Page 29	5. GPU Memory Release
Source cue	- After usage VAO and VBO should be deleted from GPU memory / - <code>glDeleteBuffers(...);</code> / - <code>glDeleteVertexArrays(...);</code>
Professor-style explanation	The slide '5. GPU Memory Release' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - After usage VAO and VBO should be deleted from GPU memory / - <code>glDeleteBuffers(...);</code> / - <code>glDeleteVertexArrays(...)</code> . The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about 5. GPU Memory Release. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - After usage VAO and VBO should be deleted from GPU memory / - <code>glDeleteBuffers(...);</code> / - <code>glDeleteVertexArrays(...);</code>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '5. GPU Memory Release'?

Page 30	Rendering Initialization 1/2
Source cue	<pre>static void init_render_scene(void) { / static const GLfloat vertices[6][2] = { / {0.0f, 1.0f}, {1.0f, 1.0f}, {1.0f, 0.0f}, /* 1st triangle */ / {0.5f, 0.5f}, {1.0f, 0.3f}, {0.7f, 1.0f} /* 2nd triangle */ / static const char* vs_src =</pre>

Professor-style explanation	The slide 'Rendering Initialization 1/2' explains the conversion from continuous geometry to a discrete sample grid. A mathematical line, triangle, or region is tested against pixel/sample positions. Covered samples become fragments, and attributes such as depth, color, normals, or texture coordinates can be interpolated across the primitive. On the slide, the concrete items are: static void init_render_scene(void) { / static const GLfloat vertices[6][2] = { / {0.0f, 1.0f}, {1.0f, 1.0f}, {1.0f, 0.0f}, /* 1st triangle */ / {0.5f, 0.5f}, {1.0f, 0.3f}, {0.7f, 1.0f} /* 2nd triangle */ / static const char* vs_src = . Rasterization creates fragment candidates; later tests decide whether those candidates become visible pixel updates.
Technical commentary	This slide is about Rendering Initialization 1/2. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: static void init_render_scene(void) { / static const GLfloat vertices[6][2] = { / {0.0f, 1.0f}, {1.0f, 1.0f}, {1.0f, 0.0f}, /* 1st triangle */ / {0.5f, 0.5f}, {1.0f, 0.3f}, {0.7f, 1.0f} /* 2nd triangle */ / static const char* vs_src =
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Rendering Initialization 1/2'?
Page 31	Rendering Initialization 2/2
Source cue	program = create_program(vs_src, fs_src); / glGenVertexArrays(1, &vao); / glGenBuffers(1, &vbo); / glBindVertexArray(vao); / glBindBuffer(GL_ARRAY_BUFFER, vbo);
Professor-style explanation	The slide 'Rendering Initialization 2/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: program = create_program(vs_src, fs_src); / glGenVertexArrays(1, &vao); / glGenBuffers(1, &vbo); / glBindVertexArray(vao); / glBindBuffer(GL_ARRAY_BUFFER, vbo). The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Rendering Initialization 2/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: program = create_program(vs_src, fs_src); / glGenVertexArrays(1, &vao); / glGenBuffers(1, &vbo); / glBindVertexArray(vao); / glBindBuffer(GL_ARRAY_BUFFER, vbo);
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Rendering Initialization 2/2'?
Page 32	Rendering

Source cue	<pre>static void render_scene(void) { / glClear(GL_COLOR_BUFFER_BIT); / glUseProgram(program); / glBindVertexArray(vao); / glDrawArrays(GL_TRIANGLES, 0, 6);</pre>
Professor-style explanation	The slide 'Rendering' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: <pre>static void render_scene(void) { / glClear(GL_COLOR_BUFFER_BIT); / glUseProgram(program); / glBindVertexArray(vao); / glDrawArrays(GL_TRIANGLES, 0, 6).</pre> The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Rendering. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: <pre>static void render_scene(void) { / glClear(GL_COLOR_BUFFER_BIT); / glUseProgram(program); / glBindVertexArray(vao); / glDrawArrays(GL_TRIANGLES, 0, 6);</pre>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Rendering'?

Page 33	Additional Attributes
Source cue	- Additional attributes are necessary for drawing / - Color, normals, texture coordinates, ... / - There are two ways to assign these attributes to the vertex data / - Attributes are bound to the same VAO in separate VBO / - Attributes are stored interleaved with vertex data,
Professor-style explanation	The slide 'Additional Attributes' explains how geometric objects change coordinate systems. A point, vector, normal, or local coordinate frame is multiplied by a transformation matrix or affected by an affine operation. Translations move positions, rotations change orientation, scaling changes size, and composed matrices combine several such effects. On the slide, the concrete items are: - Additional attributes are necessary for drawing / - Color, normals, texture coordinates, ... / - There are two ways to assign these attributes to the vertex data / - Attributes are bound to the same VAO in separate VBO / - Attributes are stored interleaved with vertex data,. The object-level relation is input coordinate space, transformation object, and output coordinate space.
Technical commentary	This slide is about Additional Attributes. The transformation objects on the slide move points, vectors, or coordinate frames from one space into another using matrices or affine operations. Concrete items shown: - Additional attributes are necessary for drawing / - Color, normals, texture coordinates, ... / - There are two ways to assign these attributes to the vertex data / - Attributes are bound to the same VAO in separate VBO / - Attributes are stored interleaved with vertex data,
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.

Check yourself	Can you state the coordinate space before and after 'Additional Attributes'?
Page 34	2.4 OpenGL Rendering Pipeline
Source cue	Programmable and static stages
Professor-style explanation	The slide '2.4 OpenGL Rendering Pipeline' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: Programmable and static stages. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about 2.4 OpenGL Rendering Pipeline. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: Programmable and static stages
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for '2.4 OpenGL Rendering Pipeline' in the rendering pipeline?
Page 35	The Rendering Pipeline 1/2
Source cue	Scene Description Raster Image / Geometry Fragment / Rasterization / Processing Processing / Vertices Primitives Fragments Pixels
Professor-style explanation	The slide 'The Rendering Pipeline 1/2' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: Scene Description Raster Image / Geometry Fragment / Rasterization / Processing Processing / Vertices Primitives Fragments Pixels. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about The Rendering Pipeline 1/2. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: Scene Description Raster Image / Geometry Fragment / Rasterization / Processing Processing / Vertices Primitives Fragments Pixels
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'The Rendering Pipeline 1/2' in the rendering pipeline?
Page 36	The Rendering Pipeline 2/2

Source cue	- Sequence of fixed and programmable stages Programmable / Vertex Processing / - Process is influenced by context and scene objects / Context Fixed Function / - Programmable levels are programmed
Professor-style explanation	The slide 'The Rendering Pipeline 2/2' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: - Sequence of fixed and programmable stages Programmable / Vertex Processing / - Process is influenced by context and scene objects / Context Fixed Function / - Programmable levels are programmed. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about The Rendering Pipeline 2/2. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: - Sequence of fixed and programmable stages Programmable / Vertex Processing / - Process is influenced by context and scene objects / Context Fixed Function / - Programmable levels are programmed
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'The Rendering Pipeline 2/2' in the rendering pipeline?

Page 37	Shader Programming – History 1/2
Source cue	- There is special graphics hardware (dedicated) / and general graphics hardware (general purpose) / - Special graphics hardware / - Volume Rendering: VolumePro / - Game consoles: Custom NVIDIA chips in Xbox
Professor-style explanation	The slide 'Shader Programming – History 1/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - There is special graphics hardware (dedicated) / and general graphics hardware (general purpose) / - Special graphics hardware / - Volume Rendering: VolumePro / - Game consoles: Custom NVIDIA chips in Xbox. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Shader Programming – History 1/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - There is special graphics hardware (dedicated) / and general graphics hardware (general purpose) / - Special graphics hardware / - Volume Rendering: VolumePro / - Game consoles: Custom NVIDIA chips in Xbox
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.

Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Programming – History 1/2'?
Page 38	Shader Programming – History 2/2
Source cue	- Various vendor-dependent extensions are introduced / to provide extensibility for general graphics hardware / - Texture shaders and register combiners allowed modification / through the application of textures / - Applicability is very limited by the possibilities of graphics hardware
Professor-style explanation	The slide 'Shader Programming – History 2/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Various vendor-dependent extensions are introduced / to provide extensibility for general graphics hardware / - Texture shaders and register combiners allowed modification / through the application of textures / - Applicability is very limited by the possibilities of graphics hardware. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Shader Programming – History 2/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Various vendor-dependent extensions are introduced / to provide extensibility for general graphics hardware / - Texture shaders and register combiners allowed modification / through the application of textures / - Applicability is very limited by the possibilities of graphics hardware
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Programming – History 2/2'?
Page 39	Assembler Shader
Source cue	- First OpenGL extensions support shader programming in assembler code / !! ARBvp1.0 / ATTRIB iPos = vertex.position ; / ATTRIB iNormal = vertex.normal ; / ...
Professor-style explanation	The slide 'Assembler Shader' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - First OpenGL extensions support shader programming in assembler code / !! ARBvp1.0 / ATTRIB iPos = vertex.position ; / ATTRIB iNormal = vertex.normal ; / . The concrete relation is between API state, GPU resource, shader input, and final rendering result.

Technical commentary	This slide is about Assembler Shader. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - First OpenGL extensions support shader programming in assembler code / !! ARBvp1.0 / ATTRIB iPos = vertex.position ; / ATTRIB iNormal = vertex.normal ; / ...
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Assembler Shader'?

Page 40	High-Level Shading Languages 1/2
Source cue	- Assembler code is hard to read and hard to maintain / - High-level languages for shading are specified in C-like syntax / - Source code is translated by the compiler into appropriate ASM statements / - Using shaders after initial binding automatically when rendering geometry / - Interface to OpenGL by passing appropriate variables or changing the OpenGL
Professor-style explanation	The slide 'High-Level Shading Languages 1/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Assembler code is hard to read and hard to maintain / - High-level languages for shading are specified in C-like syntax / - Source code is translated by the compiler into appropriate ASM statements / - Using shaders after initial binding automatically when rendering geometry / - Interface to OpenGL by passing appropriate variables or changing the OpenGL. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about High-Level Shading Languages 1/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Assembler code is hard to read and hard to maintain / - High-level languages for shading are specified in C-like syntax / - Source code is translated by the compiler into appropriate ASM statements / - Using shaders after initial binding automatically when rendering geometry / - Interface to OpenGL by passing appropriate variables or changing the OpenGL
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'High-Level Shading Languages 1/2'?

Page 41	High-Level Shading Languages 2/2
Source cue	- Existing high level languages for shader development / - CG - C for Graphics by NVIDIA / - CG was introduced in 2003 by NVIDIA as a platform independent solution / (in practice rather semi-platform independent) / - HLSL - High Level Shading Language from Microsoft

Professor-style explanation	The slide 'High-Level Shading Languages 2/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Existing high level languages for shader development / - CG - C for Graphics by NVIDIA / - CG was introduced in 2003 by NVIDIA as a platform independent solution / (in practice rather semi-platform independent) / - HLSL - High Level Shading Language from Microsoft. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about High-Level Shading Languages 2/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Existing high level languages for shader development / - CG - C for Graphics by NVIDIA / - CG was introduced in 2003 by NVIDIA as a platform independent solution / (in practice rather semi-platform independent) / - HLSL - High Level Shading Language from Microsoft
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'High-Level Shading Languages 2/2'?

Page 42	Shader Types
Source cue	- Pipeline concept has been kept in place / and shaders are responsible for different stages / - First, geometry level programming was done by vertex shaders / - Vertex shaders allow modifiability of vertex transformations and lighting / - After that, there was an extension for image-based algorithms by fragment
Professor-style explanation	The slide 'Shader Types' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Pipeline concept has been kept in place / and shaders are responsible for different stages / - First, geometry level programming was done by vertex shaders / - Vertex shaders allow modifiability of vertex transformations and lighting / - After that, there was an extension for image-based algorithms by fragment. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Shader Types. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Pipeline concept has been kept in place / and shaders are responsible for different stages / - First, geometry level programming was done by vertex shaders / - Vertex shaders allow modifiability of vertex transformations and lighting / - After that, there was an extension for image-based algorithms by fragment

Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Types'?
Page 43	Vertex Shader Concepts
Source cue	- Vertex shaders enable the programmable modification of processed geometry / - Possibilities of a vertex shader / Programmable / - Geometric transformation of vertices and normals / Vertex Processing
Professor-style explanation	The slide 'Vertex Shader Concepts' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Vertex shaders enable the programmable modification of processed geometry / - Possibilities of a vertex shader / Programmable / - Geometric transformation of vertices and normals / Vertex Processing. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Vertex Shader Concepts. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Vertex shaders enable the programmable modification of processed geometry / - Possibilities of a vertex shader / Programmable / - Geometric transformation of vertices and normals / Vertex Processing
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Concepts'?
Page 44	Vertex Shader Examples 1/2
Source cue	No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content.
Professor-style explanation	The slide 'Vertex Shader Examples 1/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Vertex Shader Examples 1/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content.
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.

Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Examples 1/2'?
Page 45	Vertex Shader Examples 2/2
Source cue	- Morphing / - Shadow volume calculation for shadow effects / [ID Software's Doom III]
Professor-style explanation	The slide 'Vertex Shader Examples 2/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Morphing / - Shadow volume calculation for shadow effects / [ID Software's Doom III]. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Vertex Shader Examples 2/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Morphing / - Shadow volume calculation for shadow effects / [ID Software's Doom III]
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Examples 2/2'?
Page 46	Vertex Shader Coordinate Systems 1/2
Source cue	- Vertex shaders are executed for each vertex / - Vertex shaders calculate coordinates in clipping coordinates as output / - Vertex shaders can not / - Access adjacent vertices / - Change the number of vertices (i.e., generate or discard vertices)
Professor-style explanation	The slide 'Vertex Shader Coordinate Systems 1/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Vertex shaders are executed for each vertex / - Vertex shaders calculate coordinates in clipping coordinates as output / - Vertex shaders can not / - Access adjacent vertices / - Change the number of vertices (i.e., generate or discard vertices). The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Vertex Shader Coordinate Systems 1/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Vertex shaders are executed for each vertex / - Vertex shaders calculate coordinates in clipping coordinates as output / - Vertex shaders can not / - Access adjacent vertices / - Change the number of vertices (i.e., generate or discard vertices)

Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Coordinate Systems 1/2'?
Page 47	Vertex Shader Coordinate Systems 2/2
Source cue	Local Space World Space View Space Clip Space Screen Space / Viewport / ■ ■ ■ / ■■■■■■ ■■■■ ■■■■■■■■ Transform
Professor-style explanation	The slide 'Vertex Shader Coordinate Systems 2/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: Local Space World Space View Space Clip Space Screen Space / Viewport / ■ ■ ■ / ■■■■■■ ■■■■ ■■■■■■■■ Transform. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Vertex Shader Coordinate Systems 2/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: Local Space World Space View Space Clip Space Screen Space / Viewport / ■ ■ ■ / ■■■■■■ ■■■■ ■■■■■■■■ Transform
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Coordinate Systems 2/2'?
Page 48	Local Space
Source cue	- Coordinates of your object relative to its local origin / - 3D vertex positions
Professor-style explanation	The slide 'Local Space' explains how geometric objects change coordinate systems. A point, vector, normal, or local coordinate frame is multiplied by a transformation matrix or affected by an affine operation. Translations move positions, rotations change orientation, scaling changes size, and composed matrices combine several such effects. On the slide, the concrete items are: - Coordinates of your object relative to its local origin / - 3D vertex positions. The object-level relation is input coordinate space, transformation object, and output coordinate space.
Technical commentary	This slide is about Local Space. The transformation objects on the slide move points, vectors, or coordinate frames from one space into another using matrices or affine operations. Concrete items shown: - Coordinates of your object relative to its local origin / - 3D vertex positions
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Local Space'?
Page 49	World Space

Source cue	- Coordinates in respect of a larger world / - Relative to some global origin of the world / - Together with many other objects
Professor-style explanation	The slide 'World Space' explains how geometric objects change coordinate systems. A point, vector, normal, or local coordinate frame is multiplied by a transformation matrix or affected by an affine operation. Translations move positions, rotations change orientation, scaling changes size, and composed matrices combine several such effects. On the slide, the concrete items are: - Coordinates in respect of a larger world / - Relative to some global origin of the world / - Together with many other objects. The object-level relation is input coordinate space, transformation object, and output coordinate space.
Technical commentary	This slide is about World Space. The transformation objects on the slide move points, vectors, or coordinate frames from one space into another using matrices or affine operations. Concrete items shown: - Coordinates in respect of a larger world / - Relative to some global origin of the world / - Together with many other objects
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'World Space'?

Page 50	View Space
Source cue	- Coordinates as seen from the camera
Professor-style explanation	The slide 'View Space' explains how geometric objects change coordinate systems. A point, vector, normal, or local coordinate frame is multiplied by a transformation matrix or affected by an affine operation. Translations move positions, rotations change orientation, scaling changes size, and composed matrices combine several such effects. On the slide, the concrete items are: - Coordinates as seen from the camera. The object-level relation is input coordinate space, transformation object, and output coordinate space.
Technical commentary	This slide is about View Space. The transformation objects on the slide move points, vectors, or coordinate frames from one space into another using matrices or affine operations. Concrete items shown: - Coordinates as seen from the camera
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'View Space'?

Page 51	Clip Space
Source cue	- Clip coordinates range from -1 to 1 / - Projection (Orthographic, Perspective)
Professor-style explanation	The slide 'Clip Space' explains how geometric objects change coordinate systems. A point, vector, normal, or local coordinate frame is multiplied by a transformation matrix or affected by an affine operation. Translations move positions, rotations change orientation, scaling changes size, and composed matrices combine several such effects. On the slide, the concrete items are: - Clip coordinates range from -1 to 1 / - Projection (Orthographic, Perspective). The object-level relation is input coordinate space, transformation object, and output coordinate space.

Technical commentary	This slide is about Clip Space. The transformation objects on the slide move points, vectors, or coordinate frames from one space into another using matrices or affine operations. Concrete items shown: - Clip coordinates range from -1 to 1 / - Projection (Orthographic, Perspective)
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Clip Space'?
Page 52	Screen Space
Source cue	- Transform to image coordinates with glViewport / - Results are sent to the rasterizer
Professor-style explanation	The slide 'Screen Space' explains how geometric objects change coordinate systems. A point, vector, normal, or local coordinate frame is multiplied by a transformation matrix or affected by an affine operation. Translations move positions, rotations change orientation, scaling changes size, and composed matrices combine several such effects. On the slide, the concrete items are: - Transform to image coordinates with glViewport / - Results are sent to the rasterizer. The object-level relation is input coordinate space, transformation object, and output coordinate space.
Technical commentary	This slide is about Screen Space. The transformation objects on the slide move points, vectors, or coordinate frames from one space into another using matrices or affine operations. Concrete items shown: - Transform to image coordinates with glViewport / - Results are sent to the rasterizer
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Screen Space'?
Page 53	Coordinate Systems
Source cue	Local Space World Space View Space Clip Space Screen Space / Viewport / ■ ■ ■ / ■■■■■ ■■■■ ■■■■■■■■ Transform
Professor-style explanation	The slide 'Coordinate Systems' explains how geometric objects change coordinate systems. A point, vector, normal, or local coordinate frame is multiplied by a transformation matrix or affected by an affine operation. Translations move positions, rotations change orientation, scaling changes size, and composed matrices combine several such effects. On the slide, the concrete items are: Local Space World Space View Space Clip Space Screen Space / Viewport / ■ ■ ■ / ■■■■■ ■■■■ ■■■■■■■■ Transform. The object-level relation is input coordinate space, transformation object, and output coordinate space.
Technical commentary	This slide is about Coordinate Systems. The transformation objects on the slide move points, vectors, or coordinate frames from one space into another using matrices or affine operations. Concrete items shown: Local Space World Space View Space Clip Space Screen Space / Viewport / ■ ■ ■ / ■■■■■ ■■■■ ■■■■■■■■ Transform
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.

Check yourself	Can you state the coordinate space before and after 'Coordinate Systems'?
Page 54	Vertex Shader Variables
Source cue	- Input types / - Uniforms: read only in VS (& FS) // Intrinsic vertex attributes / out gl_PerVertex { / - For example: light position or light color / vec4 gl_Position;
Professor-style explanation	The slide 'Vertex Shader Variables' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Input types / - Uniforms: read only in VS (& FS) // Intrinsic vertex attributes / out gl_PerVertex { / - For example: light position or light color / vec4 gl_Position. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Vertex Shader Variables. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Input types / - Uniforms: read only in VS (& FS) // Intrinsic vertex attributes / out gl_PerVertex { / - For example: light position or light color / vec4 gl_Position;
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Variables'?
Page 55	Fixed Function Vertex Postprocessing
Source cue	- Clipping / - Discard invisible geometries / Programmable / - Perspective Division / Vertex Processing
Professor-style explanation	The slide 'Fixed Function Vertex Postprocessing' explains how camera geometry maps 3D positions toward a 2D image. The projection matrix maps view-space objects into clip space. Perspective projection uses the homogeneous component so that the perspective divide makes distant objects appear smaller; orthographic projection preserves apparent size. On the slide, the concrete items are: - Clipping / - Discard invisible geometries / Programmable / - Perspective Division / Vertex Processing. The object-level chain is camera/view volume, projection matrix, clip coordinates, normalized device coordinates, and viewport coordinates.
Technical commentary	This slide is about Fixed Function Vertex Postprocessing. The projection objects on the slide convert view-space geometry into clip coordinates, normalized coordinates, and finally screen-related positions. Concrete items shown: - Clipping / - Discard invisible geometries / Programmable / - Perspective Division / Vertex Processing
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Fixed Function Vertex Postprocessing' changes positions before rasterization?

Page 56	Fixed Function Rasterization
Source cue	- Geometry is converted to raster coordinates / Programmable / Vertex Processing / Fixed Function / Primitives Fragments Vertex Postprocessing
Professor-style explanation	The slide 'Fixed Function Rasterization' explains how geometric objects change coordinate systems. A point, vector, normal, or local coordinate frame is multiplied by a transformation matrix or affected by an affine operation. Translations move positions, rotations change orientation, scaling changes size, and composed matrices combine several such effects. On the slide, the concrete items are: - Geometry is converted to raster coordinates / Programmable / Vertex Processing / Fixed Function / Primitives Fragments Vertex Postprocessing. The object-level relation is input coordinate space, transformation object, and output coordinate space.
Technical commentary	This slide is about Fixed Function Rasterization. The transformation objects on the slide move points, vectors, or coordinate frames from one space into another using matrices or affine operations. Concrete items shown: - Geometry is converted to raster coordinates / Programmable / Vertex Processing / Fixed Function / Primitives Fragments Vertex Postprocessing
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Fixed Function Rasterization'?
Page 57	Fragment Shader Concept 1/2
Source cue	- Fragment shaders set the color of a pixel / - Possibilities of a fragment shader / Programmable / - Glare of textures / Vertex Processing
Professor-style explanation	The slide 'Fragment Shader Concept 1/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Fragment shaders set the color of a pixel / - Possibilities of a fragment shader / Programmable / - Glare of textures / Vertex Processing. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Fragment Shader Concept 1/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Fragment shaders set the color of a pixel / - Possibilities of a fragment shader / Programmable / - Glare of textures / Vertex Processing
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Fragment Shader Concept 1/2'?
Page 58	Fragment Shader Concept 2/2

Source cue	- Fragment shaders are executed for fragments (= pixel predecessors) / - Fragment shaders calculate the color and depth value of the current fragment / - No information about adjacent fragments available / - The number of fragments can be changed / - Fragment shaders can not generate fragments
Professor-style explanation	The slide 'Fragment Shader Concept 2/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Fragment shaders are executed for fragments (= pixel predecessors) / - Fragment shaders calculate the color and depth value of the current fragment / - No information about adjacent fragments available / - The number of fragments can be changed / - Fragment shaders can not generate fragments. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Fragment Shader Concept 2/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Fragment shaders are executed for fragments (= pixel predecessors) / - Fragment shaders calculate the color and depth value of the current fragment / - No information about adjacent fragments available / - The number of fragments can be changed / - Fragment shaders can not generate fragments
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Fragment Shader Concept 2/2'?

Page 59	Shader Example
Source cue	layout (std140) uniform Matrices { Vertex Shader / mat4 projModelViewMatrix; / mat3 normalMatrix; /** Fragment Shader / }; * Set fragment color to red. / in vec3 position;
Professor-style explanation	The slide 'Shader Example' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: layout (std140) uniform Matrices { Vertex Shader / mat4 projModelViewMatrix; / mat3 normalMatrix; /** Fragment Shader / }; * Set fragment color to red. / in vec3 position. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Shader Example. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: layout (std140) uniform Matrices { Vertex Shader / mat4 projModelViewMatrix; / mat3 normalMatrix; /** Fragment Shader / }; * Set fragment color to red. / in vec3 position;

Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Example'?
Page 60	GLSL Shader Functions
Source cue	- Trigonometry functions / - radians, degrees, sin, cos, tan, asin, acos, atan / - Power calculations / - pow, exp2, log2, sqrt, inversesqrt / - General
Professor-style explanation	The slide 'GLSL Shader Functions' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Trigonometry functions / - radians, degrees, sin, cos, tan, asin, acos, atan / - Power calculations / - pow, exp2, log2, sqrt, inversesqrt / - General. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about GLSL Shader Functions. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Trigonometry functions / - radians, degrees, sin, cos, tan, asin, acos, atan / - Power calculations / - pow, exp2, log2, sqrt, inversesqrt / - General
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'GLSL Shader Functions'?
Page 61	Shader Usage
Source cue	- One or multiple shaders are attached to a shader program / Vertex Shader / glCreateShader glCreateProgram / Fragment Shader / glShaderSource glAttachShader glCreateShader
Professor-style explanation	The slide 'Shader Usage' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - One or multiple shaders are attached to a shader program / Vertex Shader / glCreateShader glCreateProgram / Fragment Shader / glShaderSource glAttachShader glCreateShader. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Shader Usage. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - One or multiple shaders are attached to a shader program / Vertex Shader / glCreateShader glCreateProgram / Fragment Shader / glShaderSource glAttachShader glCreateShader

Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Usage'?
Page 62	Binding Shaders
Source cue	<pre>v = glCreateShader(GL_VERTEX_SHADER); // vertex shader / char* vs = readTextFile("vertex-shader.vert"); / const char* vv = vs; / glShaderSource(v, 1, &vv, NULL); / free(vs);</pre>
Professor-style explanation	The slide 'Binding Shaders' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: <code>v = glCreateShader(GL_VERTEX_SHADER); // vertex shader / char* vs = readTextFile("vertex-shader.vert"); / const char* vv = vs; / glShaderSource(v, 1, &vv, NULL); / free(vs)</code> . The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Binding Shaders. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: <code>v = glCreateShader(GL_VERTEX_SHADER); // vertex shader / char* vs = readTextFile("vertex-shader.vert"); / const char* vv = vs; / glShaderSource(v, 1, &vv, NULL); / free(vs);</code>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Binding Shaders'?
Page 63	Setting Uniforms
Source cue	<pre>- Uniforms are used to communicate between CPU and GPU / - Data is bound to uniform locations / - Must be updated after each linking process / GLint loc0 = glGetUniformLocation(p, "isoValue_"); / glUniform1f(loc0, 0.45f);</pre>
Professor-style explanation	The slide 'Setting Uniforms' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: <code>- Uniforms are used to communicate between CPU and GPU / - Data is bound to uniform locations / - Must be updated after each linking process / GLint loc0 = glGetUniformLocation(p, "isoValue_"); / glUniform1f(loc0, 0.45f)</code> . The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.

Technical commentary	This slide is about Setting Uniforms. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: - Uniforms are used to communicate between CPU and GPU / - Data is bound to uniform locations / - Must be updated after each linking process / GLint loc0 = glGetUniformLocation(p, "isoValue_"); / glUniform1f(loc0, 0.45f);
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Setting Uniforms' into a causal sentence instead of repeating the slide title?
Page 64	Shader Performance
Source cue	- Guidelines for efficient shaders / - Since fragment shaders are executed for a large number of fragments, / they should contain as few instructions as possible / - There should be as little as possible a change of the active shaders, / because they have their own state and, accordingly, the pipeline may need
Professor-style explanation	The slide 'Shader Performance' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Guidelines for efficient shaders / - Since fragment shaders are executed for a large number of fragments, / they should contain as few instructions as possible / - There should be as little as possible a change of the active shaders, / because they have their own state and, accordingly, the pipeline may need. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Shader Performance. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Guidelines for efficient shaders / - Since fragment shaders are executed for a large number of fragments, / they should contain as few instructions as possible / - There should be as little as possible a change of the active shaders, / because they have their own state and, accordingly, the pipeline may need
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Performance'?
Page 65	Shader Development Environments 1/2
Source cue	- FX Composer, NVPerfHUD (NVIDIA) / - HLSL Shader IDE with performance analytics / www.developer.nvidia.com/page/tools.html / - RenderMonkey (ATI) / - HLSL, GLSL Shader IDE with performance analytics

Professor-style explanation	The slide 'Shader Development Environments 1/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - FX Composer, NVPerfHUD (NVIDIA) / - HLSL Shader IDE with performance analytics / www.developer.nvidia.com/page/tools.html / - RenderMonkey (ATI) / - HLSL, GLSL Shader IDE with performance analytics. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Shader Development Environments 1/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - FX Composer, NVPerfHUD (NVIDIA) / - HLSL Shader IDE with performance analytics / www.developer.nvidia.com/page/tools.html / - RenderMonkey (ATI) / - HLSL, GLSL Shader IDE with performance analytics
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Development Environments 1/2'?
Page 66	Shader Development Environments 2/2
Source cue	- Shdr online GLSL editor: https://shdr.bkcore.com/
Professor-style explanation	The slide 'Shader Development Environments 2/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Shdr online GLSL editor: https://shdr.bkcore.com/ . The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Shader Development Environments 2/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Shdr online GLSL editor: https://shdr.bkcore.com/
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Development Environments 2/2'?
Page 67	2.5 Fragment Tests and Operations
Source cue	Towards the finishing line to the pixel

Professor-style explanation	The slide '2.5 Fragment Tests and Operations' explains the conversion from continuous geometry to a discrete sample grid. A mathematical line, triangle, or region is tested against pixel/sample positions. Covered samples become fragments, and attributes such as depth, color, normals, or texture coordinates can be interpolated across the primitive. On the slide, the concrete items are: Towards the finishing line to the pixel. Rasterization creates fragment candidates; later tests decide whether those candidates become visible pixel updates.
Technical commentary	This slide is about 2.5 Fragment Tests and Operations. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: Towards the finishing line to the pixel
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by '2.5 Fragment Tests and Operations'?

Page 68	Fragment Tests and Operations 1/2
Source cue	- After fragments have been generated, they must go through the per-fragment / tests and operations / Programmable / - Only if all these tests are successful, / Vertex Processing
Professor-style explanation	The slide 'Fragment Tests and Operations 1/2' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: - After fragments have been generated, they must go through the per-fragment / tests and operations / Programmable / - Only if all these tests are successful, / Vertex Processing. The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Fragment Tests and Operations 1/2. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: - After fragments have been generated, they must go through the per-fragment / tests and operations / Programmable / - Only if all these tests are successful, / Vertex Processing
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Fragment Tests and Operations 1/2' into a causal sentence instead of repeating the slide title?

Page 69	Fragment Tests and Operations 2/2
Source cue	Fragment Pixel Ownership Scissor Alpha / Data Test Test Test / Depth Stencil / Framebuffer Dithering Blending / Test Test

Professor-style explanation	The slide 'Fragment Tests and Operations 2/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: Fragment Pixel Ownership Scissor Alpha / Data Test Test Test / Depth Stencil / Framebuffer Dithering Blending / Test Test. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Fragment Tests and Operations 2/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: Fragment Pixel Ownership Scissor Alpha / Data Test Test Test / Depth Stencil / Framebuffer Dithering Blending / Test Test
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Fragment Tests and Operations 2/2'?

Page 70	Alpha Test
Source cue	- If framebuffer is an RGBA buffer, the alpha test can be performed / - The test must first be activated / glEnable(GL_ALPHA_TEST); / - The comparison function indicates which fragments survive the test / glAlphaFunc(func, value);
Professor-style explanation	The slide 'Alpha Test' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - If framebuffer is an RGBA buffer, the alpha test can be performed / - The test must first be activated / glEnable(GL_ALPHA_TEST); / - The comparison function indicates which fragments survive the test / glAlphaFunc(func, value). The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Alpha Test. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - If framebuffer is an RGBA buffer, the alpha test can be performed / - The test must first be activated / glEnable(GL_ALPHA_TEST); / - The comparison function indicates which fragments survive the test / glAlphaFunc(func, value);
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Alpha Test'?

Page 71	Stencil Test
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Source cue	- Stencil Buffer / - Is a frame buffer component / - Is (indirectly) filled by drawing operations / - No special stencil drawing operations / - Stencil operation
Professor-style explanation	The slide 'Stencil Test' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Stencil Buffer / - Is a frame buffer component / - Is (indirectly) filled by drawing operations / - No special stencil drawing operations / - Stencil operation. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Stencil Test. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Stencil Buffer / - Is a frame buffer component / - Is (indirectly) filled by drawing operations / - No special stencil drawing operations / - Stencil operation
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Stencil Test'?

Page 72	Depth Test
Source cue	- If the framebuffer has a depth buffer, the depth test can be performed / - The test must first be activated / glEnable(GL_DEPTH_TEST); / - The comparison is made with the depth value already in the buffer / - The comparison function indicates which fragments survive the test
Professor-style explanation	The slide 'Depth Test' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - If the framebuffer has a depth buffer, the depth test can be performed / - The test must first be activated / glEnable(GL_DEPTH_TEST); / - The comparison is made with the depth value already in the buffer / - The comparison function indicates which fragments survive the test. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Depth Test. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - If the framebuffer has a depth buffer, the depth test can be performed / - The test must first be activated / glEnable(GL_DEPTH_TEST); / - The comparison is made with the depth value already in the buffer / - The comparison function indicates which fragments survive the test
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.

Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Depth Test'?
Page 73	Image Composition and Mixture
Source cue	- Alpha values of the pixels / - control the combination of pixel values / - are specified as color components of color pixels / - $\mathbf{c} = [r, g, b, a]$, $\mathbf{c} \in [0,1]$ / - Applications are manifold
Professor-style explanation	The slide 'Image Composition and Mixture' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: - Alpha values of the pixels / - control the combination of pixel values / - are specified as color components of color pixels / - $\mathbf{c} = [r, g, b, a]$, $\mathbf{c} \in [0,1]$ / - Applications are manifold. The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Image Composition and Mixture. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: - Alpha values of the pixels / - control the combination of pixel values / - are specified as color components of color pixels / - $\mathbf{c} = [r, g, b, a]$, $\mathbf{c} \in [0,1]$ / - Applications are manifold
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Image Composition and Mixture' into a causal sentence instead of repeating the slide title?
Page 74	Blending Application
Source cue	- Example application for transparency: drawing with semi-transparent brush
Professor-style explanation	The slide 'Blending Application' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: - Example application for transparency: drawing with semi-transparent brush. The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Blending Application. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: - Example application for transparency: drawing with semi-transparent brush
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Blending Application' into a causal sentence instead of repeating the slide title?
Page 75	Blending Process 1/2

Source cue	- Blending is an operation called per fragment / - Fragments are modified after rasterization, / but before writing to the framebuffer with a blending function / - Combination of a source fragment to be written with the destination pixel / in the frame buffer by using a blending function
Professor-style explanation	The slide 'Blending Process 1/2' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Blending is an operation called per fragment / - Fragments are modified after rasterization, / but before writing to the framebuffer with a blending function / - Combination of a source fragment to be written with the destination pixel / in the frame buffer by using a blending function. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Blending Process 1/2. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Blending is an operation called per fragment / - Fragments are modified after rasterization, / but before writing to the framebuffer with a blending function / - Combination of a source fragment to be written with the destination pixel / in the frame buffer by using a blending function
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Blending Process 1/2'?

Page 76	Blending Process 2/2
Source cue	- Source factor: src / - Destination factor: dst / - Source fragment: $[\text{src}, \text{src}, \text{src}, \text{src}] / \text{src}$ / - Destination pixel: $[\text{dst}, \text{dst}, \text{dst}, \text{dst}]$
Professor-style explanation	The slide 'Blending Process 2/2' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: - Source factor: src / - Destination factor: dst / - Source fragment: $[\text{src}, \text{src}, \text{src}, \text{src}] / \text{src}$ / - Destination pixel: $[\text{dst}, \text{dst}, \text{dst}, \text{dst}]$. The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Blending Process 2/2. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: - Source factor: src / - Destination factor: dst / - Source fragment: $[\text{src}, \text{src}, \text{src}, \text{src}] / \text{src}$ / - Destination pixel: $[\text{dst}, \text{dst}, \text{dst}, \text{dst}]$
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Blending Process 2/2' into a causal sentence instead of repeating the slide title?

Page 77	Applying Blending
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Source cue	- Switch on (or switch off) blending / - glEnable(GL_BLEND); / - glDisable(GL_BLEND); / - Defining Blending Factors / - glBlendFunc(srcFac, destFac);
Professor-style explanation	The slide 'Applying Blending' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: - Switch on (or switch off) blending / - glEnable(GL_BLEND); / - glDisable(GL_BLEND); / - Defining Blending Factors / - glBlendFunc(srcFac, destFac). The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Applying Blending. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: - Switch on (or switch off) blending / - glEnable(GL_BLEND); / - glDisable(GL_BLEND); / - Defining Blending Factors / - glBlendFunc(srcFac, destFac);
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Applying Blending' into a causal sentence instead of repeating the slide title?
Page 78	Blending Examples
Source cue	- Example 1: Copy source pixels and overwrite destination pixels / - glBlendFunc(GL_ONE, GL_ZERO); / - ■ = [■, ■, ■, ■] / ■■■■■■ ■,■ ■,■ ■,■ ■,■ / - Example 2: Alpha-based mixing of source pixels and destination pixels
Professor-style explanation	The slide 'Blending Examples' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: - Example 1: Copy source pixels and overwrite destination pixels / - glBlendFunc(GL_ONE, GL_ZERO); / - ■ = [■, ■, ■, ■] / ■■■■■■ ■,■ ■,■ ■,■ ■,■ / - Example 2: Alpha-based mixing of source pixels and destination pixels. The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Blending Examples. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: - Example 1: Copy source pixels and overwrite destination pixels / - glBlendFunc(GL_ONE, GL_ZERO); / - ■ = [■, ■, ■, ■] / ■■■■■■ ■,■ ■,■ ■,■ ■,■ / - Example 2: Alpha-based mixing of source pixels and destination pixels
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Blending Examples' into a causal sentence instead of repeating the slide title?
Page 79	2.6 Framebuffer
Source cue	Contains everything that is (in)visible on the screen

Professor-style explanation	The slide '2.6 Framebuffer' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: Contains everything that is (in)visible on the screen. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about 2.6 Framebuffer. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: Contains everything that is (in)visible on the screen
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '2.6 Framebuffer'?

Page 80	Viewport
Source cue	- Rectangular area in pixel grid, set explicitly / - Rendering operations are limited to this area / - All primitives that are completely or partially outside / are clipped at the viewport area (clipping)
Professor-style explanation	The slide 'Viewport' explains how camera geometry maps 3D positions toward a 2D image. The projection matrix maps view-space objects into clip space. Perspective projection uses the homogeneous component so that the perspective divide makes distant objects appear smaller; orthographic projection preserves apparent size. On the slide, the concrete items are: - Rectangular area in pixel grid, set explicitly / - Rendering operations are limited to this area / - All primitives that are completely or partially outside / are clipped at the viewport area (clipping). The object-level chain is camera/view volume, projection matrix, clip coordinates, normalized device coordinates, and viewport coordinates.
Technical commentary	This slide is about Viewport. The projection objects on the slide convert view-space geometry into clip coordinates, normalized coordinates, and finally screen-related positions. Concrete items shown: - Rectangular area in pixel grid, set explicitly / - Rendering operations are limited to this area / - All primitives that are completely or partially outside / are clipped at the viewport area (clipping)
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Viewport' changes positions before rasterization?

Page 81	Framebuffer
Source cue	- Context-bound raster with a resolution and structure / specified by the application / - Physical realization on the GPU / - Logical subdivision / - Color memory

Professor-style explanation	The slide 'Framebuffer' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Context-bound raster with a resolution and structure / specified by the application / - Physical realization on the GPU / - Logical subdivision / - Color memory. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Framebuffer. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Context-bound raster with a resolution and structure / specified by the application / - Physical realization on the GPU / - Logical subdivision / - Color memory
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Framebuffer'?

Page 82	OpenGL Framebuffers 1/3
Source cue	- Typical OpenGL framebuffer configuration / 8bit Stencil Buffer / 32bit Depth Buffer / = 32bit Color Buffer
Professor-style explanation	The slide 'OpenGL Framebuffers 1/3' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Typical OpenGL framebuffer configuration / 8bit Stencil Buffer / 32bit Depth Buffer / = 32bit Color Buffer. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about OpenGL Framebuffers 1/3. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Typical OpenGL framebuffer configuration / 8bit Stencil Buffer / 32bit Depth Buffer / = 32bit Color Buffer
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Framebuffers 1/3'?

Page 83	OpenGL Framebuffers 2/3
Source cue	- All buffers of a frame buffer have the same grid size / - Number of bitplanes per buffer can be configured individually / - Individual buffers are directly supported by the hardware / - Buffer identifier / - GL_COLOR_BUFFER_BIT

Professor-style explanation	The slide 'OpenGL Framebuffers 2/3' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - All buffers of a frame buffer have the same grid size / - Number of bitplanes per buffer can be configured individually / - Individual buffers are directly supported by the hardware / - Buffer identifier / - GL_COLOR_BUFFER_BIT. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about OpenGL Framebuffers 2/3. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - All buffers of a frame buffer have the same grid size / - Number of bitplanes per buffer can be configured individually / - Individual buffers are directly supported by the hardware / - Buffer identifier / - GL_COLOR_BUFFER_BIT
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Framebuffers 2/3'?

Page 84	OpenGL Framebuffers 3/3
Source cue	- Delete a buffer / <code>glClear(GLbitfield mask);</code> / - Set the initialization color for the color buffer / <code>glClearColor(1.0,1.0,1.0,1.0);</code> / <code>glClear(GL_COLOR_BUFFER_BIT);</code>
Professor-style explanation	The slide 'OpenGL Framebuffers 3/3' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Delete a buffer / <code>glClear(GLbitfield mask);</code> / - Set the initialization color for the color buffer / <code>glClearColor(1.0,1.0,1.0,1.0);</code> / <code>glClear(GL_COLOR_BUFFER_BIT);</code> . The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about OpenGL Framebuffers 3/3. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Delete a buffer / <code>glClear(GLbitfield mask);</code> / - Set the initialization color for the color buffer / <code>glClearColor(1.0,1.0,1.0,1.0);</code> / <code>glClear(GL_COLOR_BUFFER_BIT);</code>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Framebuffers 3/3'?

Page 85	Double Buffering
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Source cue	- Use of two color buffers to avoid flicker effects / - Image buildup takes place in the background (back buffer) / - Visible image is independent of this (front buffer) / Rendering / Process
Professor-style explanation	The slide 'Double Buffering' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Use of two color buffers to avoid flicker effects / - Image buildup takes place in the background (back buffer) / - Visible image is independent of this (front buffer) / Rendering / Process. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Double Buffering. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Use of two color buffers to avoid flicker effects / - Image buildup takes place in the background (back buffer) / - Visible image is independent of this (front buffer) / Rendering / Process
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Double Buffering'?

Page 86	2.7 Pixel-based Rendering
Source cue	Direct pixel processing without involving 3D models
Professor-style explanation	The slide '2.7 Pixel-based Rendering' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: Direct pixel processing without involving 3D models. The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about 2.7 Pixel-based Rendering. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: Direct pixel processing without involving 3D models
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn '2.7 Pixel-based Rendering' into a causal sentence instead of repeating the slide title?

Page 87	Pixel-based Rendering Operations
Source cue	- Generally / - Read and write operations for various image data formats, / usually as an external library (e.g., DevIL) / - Conversions between image data types / - In OpenGL done with framebuffer objects

Professor-style explanation	The slide 'Pixel-based Rendering Operations' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Generally / - Read and write operations for various image data formats, / usually as an external library (e.g., DevIL) / - Conversions between image data types / - In OpenGL done with framebuffer objects. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Pixel-based Rendering Operations. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Generally / - Read and write operations for various image data formats, / usually as an external library (e.g., DevIL) / - Conversions between image data types / - In OpenGL done with framebuffer objects
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Pixel-based Rendering Operations'?

Page 88	Image Operations
Source cue	- Rendering image data with OpenGL / Copy Pixel Data / Draw/Read / Pixel / Geometry
Professor-style explanation	The slide 'Image Operations' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Rendering image data with OpenGL / Copy Pixel Data / Draw/Read / Pixel / Geometry. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Image Operations. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Rendering image data with OpenGL / Copy Pixel Data / Draw/Read / Pixel / Geometry
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Image Operations'?

Page 89	Write Pixels into Framebuffer
Source cue	- glDrawPixels(x, y, width, height, format, type, pixels); / - Example / // 64x64 array with rgb components / GLubyte img[64][64][3]; / void draw() {

Professor-style explanation	The slide 'Write Pixels into Framebuffer' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - <code>glDrawPixels(x, y, width, height, format, type, pixels);</code> / - Example // 64x64 array with rgb components / <code>GLubyte img[64][64][3];</code> / <code>void draw() {</code> . The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Write Pixels into Framebuffer. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - <code>glDrawPixels(x, y, width, height, format, type, pixels);</code> / - Example // 64x64 array with rgb components / <code>GLubyte img[64][64][3];</code> / <code>void draw() {</code>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Write Pixels into Framebuffer'?

Page 90	Read Pixels from Framebuffer
Source cue	- <code>glReadPixels(x, y, width, height, format, type, pixels);</code> / - Example / <code>GLubyte* img = 0;</code> / <code>int width, height;</code> // canvas size / <code>void snapshot() {</code>
Professor-style explanation	The slide 'Read Pixels from Framebuffer' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - <code>glReadPixels(x, y, width, height, format, type, pixels);</code> / - Example / <code>GLubyte* img = 0;</code> / <code>int width, height;</code> // canvas size / <code>void snapshot() {</code> . The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Read Pixels from Framebuffer. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - <code>glReadPixels(x, y, width, height, format, type, pixels);</code> / - Example / <code>GLubyte* img = 0;</code> / <code>int width, height;</code> // canvas size / <code>void snapshot() {</code>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Read Pixels from Framebuffer'?

Page 91	Formats & Data Types
Source cue	- Format information for pixel array operations / - <code>GL_RGB</code> – red-green-blue values of the frame buffer (FB) / - <code>GL_RED</code> – red values of the FB / - <code>GL_ALPHA</code> – transparency values of the FB / - <code>GL_RGBA</code> – red-Green-Blue-Alpha values of the FB

Professor-style explanation	The slide 'Formats & Data Types' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Format information for pixel array operations / - GL_RGB – red-green-blue values of the frame buffer (FB) / - GL_RED – red values of the FB / - GL_ALPHA – transparency values of the FB / - GL_RGBA – red-Green-Blue-Alpha values of the FB. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Formats & Data Types. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Format information for pixel array operations / - GL_RGB – red-green-blue values of the frame buffer (FB) / - GL_RED – red values of the FB / - GL_ALPHA – transparency values of the FB / - GL_RGBA – red-Green-Blue-Alpha values of the FB
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Formats & Data Types'?

Page 92	coordinate systems in the geometry stage
Source cue	- Rendering pipeline consist of geometry stage and rasterization stage / - Geometric primitives (mostly triangles) are transformed through different / coordinate systems in the geometry stage / - Rasterization results in fragments, which are pixel predecessors / - Fragments have to endure a sequence of tests in the rasterization stage, in
Professor-style explanation	The slide 'coordinate systems in the geometry stage' explains object-based rendering as a conversion chain. A model supplies geometric objects such as vertices or primitives. The geometry stage transforms and assembles those objects. Rasterization turns primitives into fragments, and fragment operations decide which fragments update pixels in the framebuffer. On the slide, the concrete items are: - Rendering pipeline consist of geometry stage and rasterization stage / - Geometric primitives (mostly triangles) are transformed through different / coordinate systems in the geometry stage / - Rasterization results in fragments, which are pixel predecessors / - Fragments have to endure a sequence of tests in the rasterization stage, in. The essential relation is that every stage changes the representation of the same scene information.
Technical commentary	This slide is about coordinate systems in the geometry stage. Scene or model data moves through a sequence of representations: vertices, primitives, fragments, tests, and framebuffer updates. Each named object is one stage in that conversion. Concrete items shown: - Rendering pipeline consist of geometry stage and rasterization stage / - Geometric primitives (mostly triangles) are transformed through different / coordinate systems in the geometry stage / - Rasterization results in fragments, which are pixel predecessors / - Fragments have to endure a sequence of tests in the rasterization stage, in

Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'coordinate systems in the geometry stage' in the rendering pipeline?
Page 93	Literature and other sources used in this chapter
Source cue	No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content.
Professor-style explanation	The slide 'Literature and other sources used in this chapter' introduces a concrete graphics object, operation, or relation. The named terms describe input data, a processing step, and an output used elsewhere in the rendering workflow. On the slide, the concrete items are: No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content. The object-level relation is therefore input, operation, output, and the later graphics stage that consumes the output.
Technical commentary	This slide is about Literature and other sources used in this chapter. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Literature and other sources used in this chapter' into a causal sentence instead of repeating the slide title?
Page 94	Guide: The Official Guide to Learning OpenGL
Source cue	- Text Books / - D. Shreiner, G. Sellers, J. Kessenich, B. Licea-Kane: OpenGL Programming / Guide: The Official Guide to Learning OpenGL / (8th Edition), Addison-Wesley 2013. / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer
Professor-style explanation	The slide 'Guide: The Official Guide to Learning OpenGL' describes the OpenGL side of the rendering pipeline. OpenGL represents rendering through objects such as contexts, buffers, vertex arrays, shaders, textures, samplers, and framebuffers. At draw time, the currently bound objects and state determine what data reaches the GPU and how the pipeline processes it. On the slide, the concrete items are: - Text Books / - D. Shreiner, G. Sellers, J. Kessenich, B. Licea-Kane: OpenGL Programming / Guide: The Official Guide to Learning OpenGL / (8th Edition), Addison-Wesley 2013. / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer. The concrete relation is between API state, GPU resource, shader input, and final rendering result.
Technical commentary	This slide is about Guide: The Official Guide to Learning OpenGL. The OpenGL objects and calls shown here control GPU state, bound resources, shader interfaces, buffer contents, or framebuffer access at draw time. Concrete items shown: - Text Books / - D. Shreiner, G. Sellers, J. Kessenich, B. Licea-Kane: OpenGL Programming / Guide: The Official Guide to Learning OpenGL / (8th Edition), Addison-Wesley 2013. / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer

Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Guide: The Official Guide to Learning OpenGL'?

02.1 Rendering Process

Source location: Section 2.1

Commentary

Object-based rendering starts with scene objects represented as primitives. The pipeline transforms vertices, assembles primitives, rasterizes them into fragments, and lets only some fragments become pixels. The important detail is that the representation changes at every stage.

The application stage prepares data and state. The geometry stage handles positions and primitives. The rasterization stage creates fragments and applies fragment operations. This separation helps you explain both theory questions and black-screen OpenGL bugs.

Mental model

Vertices become primitives, primitives become fragments, and fragments compete to become pixel updates.

Check yourself

What is the difference between a fragment and a final pixel?

02.2 OpenGL Overview

Source location: Section 2.2

Commentary

OpenGL is an API for controlling a graphics pipeline. It is not a renderer by itself in the sense of a single command that understands your scene. You create a context, provide buffers, set state, compile shaders, bind objects, and issue draw calls.

Modern OpenGL is stateful and shader-based. Many errors happen because the programmer assumes that a later call carries more information than it really does. The GPU uses the currently bound objects and current state at draw time.

Mental model

OpenGL draw calls consume the state machine as it exists right now.

Check yourself

Why can binding the wrong vertex array or shader program produce a valid draw call with wrong output?

02.3 OpenGL Rendering

Source location: Section 2.3

Commentary

Older OpenGL exposed a fixed-function style where many transformations and lighting choices were configured through predefined calls. Modern OpenGL expects you to write shader programs and explicitly manage data flow.

This is not just historical trivia. It explains why shader code, vertex attributes, uniforms, and buffer objects are central in the exercises. If the shader expects an attribute and the vertex layout does not provide it, the pipeline has no magic fallback.

Mental model

Modern OpenGL makes the data path explicit; that gives control but also creates responsibility.

Check yourself

Which data usually goes into vertex attributes, and which data usually goes into uniforms?

02.4 OpenGL Rendering Pipeline

Source location: Section 2.4

Commentary

The vertex shader runs per vertex and usually outputs clip-space position plus attributes for later interpolation. Primitive assembly groups vertices into points, lines, or triangles. Clipping handles geometry outside the view volume. Rasterization creates fragments.

The fragment shader computes candidate fragment output, often using interpolated attributes, uniforms, and textures. After that, tests and operations such as depth testing, stencil testing, blending, and masking decide what reaches the framebuffer.

Mental model

Programmable shaders compute values; fixed pipeline stages decide coverage, interpolation, tests, and output rules.

Check yourself

If your geometry appears in wireframe but colors are wrong, which stages become suspicious?

02.5 Fragment Tests, Framebuffer, and Pixel-Based Rendering

Source location: Sections 2.5-2.7

Commentary

Fragment tests are the gatekeepers after shading. A fragment can have a perfectly valid color and still fail because depth, stencil, scissor, masking, or blending state prevents a visible update. This is why final pixels are a subset of generated fragments.

The framebuffer stores the final render targets: color buffers, depth buffers, stencil buffers, or custom attachments. Pixel-based rendering starts from image data instead of geometric primitives, which is useful for image processing but conceptually different from the object-to-pixel pipeline.

Mental model

The framebuffer is the destination; fragment operations are the rules for writing into it.

Check yourself

Why can two fragments with different colors produce only one visible pixel color?

End-of-lecture summary

If you remember only one thing from Lecture 02, remember this: This lecture turns the vague idea of rendering into a concrete sequence. It is the most important debugging map in the course: if an image is wrong, the pipeline tells you where the error could have entered.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.